



e-Learning through VR in the classroom (Credit: <https://elearningindustry.com>)

real environment. These use advanced processing, sensing and display technology, and are ideal for defence sector applications to gain experience of dangerous environments from the convenience of a training room. Such applications can put trainees in different situations and environments for teaching awareness and upskilling to provide valuable experiences.

Retail industry

AR and VR have the potential to revolutionise customer experience and drive revenues by individualising retailer offers and enabling customers to visualise products in different settings. Retailers can also enhance the understanding of information through advanced graphical visualisation and simulation technologies.

Pilots and implementation examples include VR tours by Tesco, Alibaba's full VR shopping experience, Adidas' VR videos to promote its outdoor clothing collection and FR Tech-iLUK for personalised styling. With AR-enabled platforms, buyers can also interact with the product(s) they want to buy to better understand them and explore their functionalities.

Business processes

If such AR displays as smart glasses become more common, identity detection can enrich everyday conversations by providing details such as projects a colleague is working on. Immersive telepresence enhancements come by adding elements of VR, AR and AI to improve the collaborative experience.

Indrajit Belgundi, general manager, Client Solutions Group, says, "AR/VR will bring more on-site learning and

creativity to the workplace. On-site training opportunities and the ability to access data in real time will give the workforce freedom to do their best work untethered from the workplace. Further, employees will be able to collaborate as if they are all physically working together, bringing everyone into one virtual environment."

Education and entertainment industries

The education sector has become the most funded industry, bringing in new-age education for the next generation in the form of app-based learning and smart classes. STEAM (Science, Technology, Engineering, Arts and Mathematics) education offers new-age experiential learning for students age six onwards, in the form of robotic kits, AI programming, AR/VR interactive classes, the Internet of Things (IoT) applications, 3D printing, STEAM activities and training programs, among others. AR and VR apps in the education sector add an interactive audio-visual factor to the e-learning process.

Platform such as Augment Education are great for making presentations, modelling and 3D designing, for constructing various e-learning materials using objects and animating them employing various shapes and actions.

Another example is SignAloud Gloves tool for students with impairments. The tool allows students to communicate using sign language in a VR environment and translates the same into human speech.

2D, 3D, 4D and interactive AR/VR content used for gaming and films

can be created through flexible and high-performance end-to-end development platforms. India has a huge potential in Industry 4.0 in terms of evolving capabilities for creating digital content, ranging from animation, gaming, VFX, AR and VR.

Healthcare

AR and VR together with AI can educate the next generation of professionals in life-threatening and responsible jobs such as doctors, firemen, factory workers, pilots and the like. Doctors can go through countless virtual scenarios of operations without putting a single life in danger. Firemen teams can go through the worst cases, pilots can bring back the plane's latitude to the haven or emergency landing, medical teams can save people's lives from traffic accidents and natural disasters.

AR and VR have positively impacted the healthcare industry just like the Internet of Things (IoT), AI and ML. There are several examples of AR and VR in the healthcare sector, from training and surgery guidance to organ-mapping in real time that help doctors avoid errors.

Future scope

AR or VR will help in the transformation of industries across the board. These technologies will continue to help achieve better designs and faster time-to-market through immersive training tools offering simulations and scenarios in a secure, risk-free environment.

Recent developments in technology that have enhanced VR capabilities are deep learning to foster real-time image and speech recognition, increased availability and reduced cost of local processing and storage, network bandwidth expansion with 5G, richer data streams and availability of AI in the cloud.

5G mobile network technology provides an opportunity to accelerate the adoption of AR and VR. Gartner expects that implementation of 5G and AR/VR in stores will transform shopping experiences and the entire product management cycle of brands.

Both AR and VR combined with AI and ML will create a better future, and take industries and businesses to an entirely new level. **EFY**